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1  # Note 01: Data Visualization Techniques
2  # =====
3
4  # Import Libraries
5  import matplotlib.pyplot as plt
6  import numpy as np
7  from scipy.stats import norm
8  import seaborn as sns
9
10 # 01. Scatter Plot
11 # =====
12
13 x = [1, 2, 3, 4, 5]
14 y = [2, 4, 5, 8, 10]
15
16 plt.figure()
17 plt.scatter(x, y, s=30, c="purple", marker="o")
18 plt.xlabel("X")
19 plt.ylabel("Y")
20 plt.title("Scatter Plot")
21 plt.grid(True, alpha=0.3)
22 plt.show()
23
24 # 02. Bar Plot
25 # =====
26
27 # 1. Simple Bar Chart - Vertical Chart
28 products = ["A", "B", "C", "D"]
29 sales = [100, 150, 80, 200]
30
31 plt.figure()
32 plt.bar(products, sales, width=0.4, color="green")
33 plt.xlabel("Products")
34 plt.ylabel("Sales")
35 plt.title("Bar Chart")
36 plt.grid(True, alpha=0.3)
37 plt.show()
38
39 # 2. Simple Bar Chart - Horizontal Chart
40 products = ["A", "B", "C", "D"]
41 sales = [100, 150, 80, 200]
42
43 plt.figure()
44 plt.barh(products, sales, height=0.4, color="yellow")
45 plt.ylabel("Products")
46 plt.xlabel("Sales")
47 plt.title("Bar Chart")
48 plt.grid(True, alpha=0.3)
49 plt.show()
50
51 # 3. Multiple Bar Chart - Vertical Chart
52 cats = ['A', 'B', 'C', 'D']
53 v1, v2 = [4, 7, 1, 8], [5, 6, 2, 9]
54 w, x = 0.4, np.arange(len(cats))
55
56 plt.figure()
57 plt.bar(x - w/2, v1, w, label='Set 1')
58 plt.bar(x + w/2, v2, w, label='Set 2')
59 plt.xticks(x, cats)
60 plt.xlabel('Cats')
61 plt.ylabel('Values')
62 plt.title('Grouped Bar Chart')
63 plt.legend()
64 plt.grid(True, alpha=0.3)
65 plt.show()
66
67 # 4. Multiple Bar Chart - Horizontal Chart
68 cats = ['A', 'B', 'C', 'D']
69 v1, v2, v3 = [4, 7, 1, 8], [5, 6, 2, 9], [9, 10, 11, 13]
70 w, x = 0.3, np.arange(len(cats))
71
72 plt.figure()

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73 plt.barh(x - w, v1, height=w, label='Set 1')
74 plt.barh(x, v2, height=w, label='Set 2')
75 plt.barh(x + w, v3, height=w, label='Set 3')
76 plt.yticks(x, cats)
77 plt.xlabel('Values')
78 plt.ylabel('Cats')
79 plt.title('Grouped Bar Chart')
80 plt.legend()
81 plt.grid(True, alpha=0.3, linestyle='--')
82 plt.show()
83
84 # 5. Stacked Bar Chart - Vertical Chart
85 x = ['A', 'B', 'C', 'D']
86 y1 = [10, 20, 10, 30]
87 y2 = [20, 25, 15, 25]
88
89 plt.figure()
90 plt.bar(x, y1, color='red')
91 plt.bar(x, y2, bottom=y1, color='b')
92 plt.grid(True, alpha=0.3)
93 plt.ylabel("Values")
94 plt.title("Bar Chart")
95 plt.xlabel("Cats")
96 plt.show()
97
98 # 6. Stacked Bar Chart - Horizontal Chart
99 y = ['A', 'B', 'C', 'D']
100 x1 = [10, 20, 10, 30]
101 x2 = [20, 25, 15, 25]
102
103 plt.figure()
104 plt.barh(y, x1, color='purple')
105 plt.barh(y, x2, left=x1, color='blue')
106 plt.grid(True, alpha=0.3)
107 plt.ylabel("Cats")
108 plt.title("Bar Chart")
109 plt.xlabel("Values")
110 plt.show()
111
112 # 03. Pie Chart
113 # =====
114
115 labels = ["IT", "Business", "Engineering", "Arts"]
116 values = [40, 25, 20, 15]
117
118 plt.figure()
119 plt.pie(values, labels=labels, autopct="%1.2f%", wedgeprops={"edgecolor": "black",
120 "linewidth": 1})
121 plt.title("Student Distribution")
122 plt.show()
123
124 # 04. Line Chart
125 # =====
126
127 months = ["Jan", "Feb", "Mar", "Apr", "May"]
128 sales = [100, 120, 150, 140, 180]
129
130 plt.figure()
131 plt.plot(months, sales, linestyle="--", color="red")
132 plt.scatter(months, sales, s=30, c="purple", marker="o", label="2021")
133 plt.xlabel("Month")
134 plt.ylabel("Sales")
135 plt.title("Monthly Sales")
136 plt.grid(True, alpha=0.3)
137 plt.legend(loc="lower right")
138 plt.show()
139
140 # Legend Locations
141 # =====
142
143 # +-----+-----+-----+
144 # | 'upper left' | 'upper center' | 'upper right' |

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144 # +-----+-----+-----+
145 # |'center left' | 'center' |'center right' |
146 # +-----+-----+-----+
147 # | 'lower left' |'lower center'| 'lower right' |
148 # +-----+-----+-----+
149
150 # 05. Histogram
151 # =====
152
153 marks = [
154     55, 60, 62, 65, 65,
155     68, 68, 70, 70, 70,
156     72, 72, 72, 72, 73,
157     73, 74, 74, 75, 75,
158     75, 76, 76, 76, 77,
159     77, 78, 78, 78, 80,
160     80, 81, 82, 82, 83,
161     84, 85, 86, 88, 90
162 ]
163
164 # Normal Distribution Curve
165 mean = np.mean(marks)
166 std = np.std(marks)
167 x = np.linspace(min(marks), max(marks), 100)
168 y = norm.pdf(x, mean, std)
169
170 # Plot Graph and Curve
171 plt.figure()
172 plt.hist(marks, bins=10, density=True, edgecolor="black", linewidth=1.5)
173 plt.plot(x, y, linewidth=2, linestyle="--", color="red")
174 plt.xlabel("Marks")
175 plt.ylabel("Density")
176 plt.title("Distribution of Marks")
177 plt.grid(True, alpha=0.3)
178 plt.show()
179
180 # 06. Boxplot
181 # =====
182
183 marks_A = [45, 50, 52, 55, 60, 62, 65, 65, 70, 72, 75, 80, 85, 95]
184 marks_B = [50, 55, 58, 60, 63, 65, 68, 70, 72, 75, 78, 80, 82, 85]
185 marks_C = [40, 45, 48, 50, 55, 58, 60, 62, 65, 68, 70, 72, 75, 78]
186
187 plt.figure()
188 plt.boxplot([marks_A, marks_B, marks_C])
189 plt.xticks([1, 2, 3], ["Class A", "Class B", "Class C"])
190 plt.ylabel("Marks")
191 plt.xlabel("Classes")
192 plt.title("Distribution of Marks")
193 plt.grid(True, alpha=0.3, linestyle='--')
194 plt.show()
195
196 # 07. Heatmap
197 # =====
198
199 data = np.random.rand(10, 10) * 100
200
201 plt.figure()
202 sns.heatmap(
203     data,
204     xticklabels=list("ABCDEFGHIJ"),
205     yticklabels=False,
206     cmap="coolwarm",
207     linewidths=0.5,
208     annot=True,
209     fmt=".1f"
210 )
211 plt.title("Heatmap", fontsize=16)
212 plt.xlabel("X-axis")
213 plt.ylabel("Y-axis")
214 plt.show()
215

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216 # 08. Area Chart
217 # =====
218
219 months = ["Jan", "Feb", "Mar", "Apr", "May"]
220 sales_2025 = [100, 120, 150, 140, 180]
221 sales_2026 = [110, 130, 160, 155, 200]
222
223 plt.figure()
224 plt.fill_between(months, sales_2025, sales_2026, alpha=0.3)
225 # plt.fill_between(months, sales_2026, alpha=0.3)
226 plt.plot(months, sales_2025, marker="o", label="2025")
227 plt.plot(months, sales_2026, marker="o", label="2026")
228 plt.xlabel("Month")
229 plt.ylabel("Sales")
230 plt.title("Sales Over Time")
231 plt.legend()
232 plt.grid(True, alpha=0.3, linestyle='--')
233 plt.show()
234
235 # 09. 3D Plot
236 # =====
237
238 # 1. 3D Scatter Plot
239 fig = plt.figure()
240 ax = fig.add_subplot(111, projection="3d")
241 # 111 => 1 Row / 1 Column / 1 Plot
242 # 3d => Prepare a 3D Canvas
243 x = [1, 2, 3, 4, 5]
244 y = [2, 4, 1, 5, 3]
245 z = [5, 3, 4, 2, 6]
246
247 ax.scatter(x, y, z)
248 ax.set_xlabel("X")
249 ax.set_ylabel("Y")
250 ax.set_zlabel("Z")
251 plt.title("3D Scatter Plot")
252 ax.grid(True, alpha=0.3)
253 plt.show()
254
255 # 2. 3D Line Plot
256 fig = plt.figure()
257 ax = fig.add_subplot(111, projection="3d")
258 x = [1, 2, 3, 4, 5]
259 y = [2, 4, 1, 5, 3]
260 z = [5, 3, 4, 2, 6]
261
262 ax.plot(x, y, z)
263 ax.set_xlabel("X")
264 ax.set_ylabel("Y")
265 ax.set_zlabel("Z")
266 plt.title("3D Line Plot")
267 plt.show()
268
269 # 3. 3D Surface Plot
270 x = np.linspace(-5, 5, 50)
271 y = np.linspace(-5, 5, 50)
272 # Meshgrid combines every x with every y
273 X, Y = np.meshgrid(x, y)
274 Z = X**2 + Y**2
275
276 fig = plt.figure()
277 ax = fig.add_subplot(111, projection="3d")
278 ax.plot_surface(X, Y, Z)
279 ax.set_xlabel("X")
280 ax.set_ylabel("Y")
281 ax.set_zlabel("Z")
282 plt.title("3D Surface Plot")
283 plt.show()
284
285 # 10. Error Bar Plot
286 # =====
287

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288 # Data
289 subjects = ['A', 'B', 'C', 'D', 'E']
290
291 # Mean values
292 mean = np.array([70, 75, 65, 80, 72])
293
294 # Error values
295 error = np.array([5, 4, 6, 3, 5])
296
297 # Create error bar plot
298 plt.errorbar(
299     subjects,
300     mean,
301     yerr=error, # Vertical Error
302     # xerr=error, # Horizontal Error
303     fmt='o',
304     capsize=5,
305     markersize=7
306 )
307
308 # Labels and title
309 plt.xlabel("Subjects")
310 plt.ylabel("Mean Marks")
311 plt.title("Mean Marks with Error Bars")
312
313 # Grid
314 plt.grid(True)
315
316 # Display plot
317 plt.show()
```